

Updating risk and protective factors for dementia in older adults

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Abstract

Promoting good cognitive functioning and preventing dementia in late life are priority actions for many countries with rapidly ageing populations. Although epidemiological research has established some modifiable factors that influence the risk of cognitive impairment (such as low education level, hearing impairment, smoking, obesity, physical inactivity or social isolation), a growing literature suggests further risk and protective factors that might affect cognitive functioning and dementia. In this Review, we examine the potential effects of these less well-established factors, and discuss promotion and prevention strategies at the population and individual levels that might reduce dementia risk in the long term. Reducing financial struggles, neighbourhood deprivation and workplace strain and promoting leisure activities, emotional wellbeing and healthy nutritional styles have emerged as factors that might help to prevent dementia and could be included among priority actions for healthy cognitive ageing.

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Introduction

Dementia is characterized by a severe level of cognitive impairment that limits a person's ability to manage life independently¹. Dementia risk increases with age². A long preclinical phase of cognitive decline usually precedes the diagnosis of dementia³, and the biological pathways that underlie normative cognitive decline in old age and dementia overlap⁴. However, the neurobiological and cognitive deterioration that characterize dementia (such as the accumulation of amyloid plaques and neurofibrillary tangles in the brain in Alzheimer disease⁵) is worse than that of normal cognitive ageing. The cognitive abilities that make up fluid intelligence (logic reasoning such as generating, transforming and manipulating different types of novel information in real time) decline with age. However, crystallized intelligence (a person's general knowledge, vocabulary and reasoning based on previously learned information) remains intact until late in life⁶. When a person experiences cognitive decline (as recognized by themselves or a proxy and confirmed in cognitive testing) but can still function independently, and if their symptoms do not meet the criteria for dementia, they are clinically classified as having 'mild cognitive impairment'⁷.

An increasing life expectancy in most countries has directed public and research attention to healthy cognitive ageing⁸ and the prevention of cognitive decline and dementia in old age⁹. Epidemiological studies have revealed numerous modifiable lifestyle factors that influence dementia risk (Box 1), of which 14 might be responsible for about 45% of dementia cases globally¹⁰ and are well established dementia risk factors¹¹: excessive alcohol consumption, head injuries, air pollution, low education level, high blood pressure, hearing impairment, untreated vision loss, smoking, obesity, high LDL cholesterol, depression, physical inactivity, diabetes and social isolation. Most of these factors are incorporated in the 'Lifestyle for Brain Health' index¹² (a dementia risk algorithm developed in an expert Delphi consensus process), which also lists sleep difficulties¹³. Insomnia, fragmentation of sleep, rapid-eye-movement sleep behaviour disorder and other types of sleep disorder were associated with higher dementia risk in a meta-analysis of 51 longitudinal studies¹⁴. By contrast, adherence to the Mediterranean diet¹⁵ and cognitive training¹⁶ are established lifestyle factors related to reduced dementia risk¹⁷.

A better knowledge of specific risk and protective factors is necessary to design public health programmes for everyone across societies. Although each currently established risk factor is relevant, other risk factors could also be important. Less well-established risk and protective lifestyle factors should be identified and integrated into promotion and prevention programmes, yielding opportunities to achieve better cognitive outcomes and reduce vulnerability. Indeed, epidemiological studies have reported associations between other less established, potentially modifiable lifestyle factors and worse cognitive functioning in older age, but evidence has not yet accumulated to the extent that they are widely recognized. For example, life-course socioeconomic inequalities and psychological distress have been proposed as risk factors, but the evidence has been judged insufficient to recommend lifestyle changes¹³.

In this Review, we describe the current state of the art of these less well-established, modifiable lifestyle factors and consider their risks and protective roles in cognitive decline and dementia in older adults. We identified these factors on the basis of epidemiological evidence published over the past two decades and grouped them into population-level and individual-level factors. We first discuss evidence for the population-level factors (financial resources, neighbourhood characteristics and the workplace) and the individual-level factors

(leisure activities, emotional wellbeing, nutritional style and lifetime diagnosis of psychiatric disorders) identified in cross-sectional or longitudinal epidemiological studies. Next, we look at the available evidence from intervention studies to explore possible causal links between dementia and the identified factors, and discuss possible targets for promotion and prevention programmes. We then examine the potential mechanisms that might underlie the identified associations. We end by considering the interactions between risk and protective factors across levels and the need for translational research to deliver effective interventions.

Population-level factors

Population-level factors are macro-level and meso-level determinants that usually cannot be influenced directly by an individual and require societal-level changes to be modified¹⁸. If population-level factors were changed, they could affect more people than individual-level factors; population-level factors can comprise available financial resources and neighbourhood or workplace characteristics.

Financial resources

Financial resources are the monetary assets available to a person. This factor was investigated in the reviewed literature in terms of income, wealth, poverty, economic resources and deprivation.

Evidence from the USA ($n = 10,781$) indicated that high late-life income was associated with slower cognitive decline¹⁹, an effect still observable among people in their 70s (Memphis and Pittsburgh sample, $n = 2,574$)²⁰. An analysis of data from 6,232 adults in the USA suggested that up to 12% of the cognitive performance disparity in old age between Black and white people was related to wealth²¹.

Concerning dementia risk, higher income was associated with lower odds of dementia among Medicare recipients in the USA ($n = 3,785$)²². Similar observations for wealth were made in a UK sample ($n = 6,346$)²³ and for poverty in studies with South African ($n = 227$) and Pakistani ($n = 817$) populations^{24,25}. Further, negative wealth shocks (suddenly losing 75% or more of one's total wealth) were associated with faster cognitive decline as well as higher dementia risk in a study in the USA ($n = 8,082$)²⁶.

Social deprivation is a lack of the resources considered standard in a given society²⁷. Analyses of European data ($n = 11,623$) using an index of individual social deprivation that included measures of income and wealth, job stability and health insurance revealed associations between higher social deprivation and a higher risk for developing dementia²⁸. An analysis of data from the USA ($n = 3,867$) showed that only 27% of the association between individual social deprivation and cognitive performance in older age was mediated by lifestyle activities such as attending educational courses, reading, volunteering or hobbies²⁹, indicating that other aspects might be responsible for differences in cognitive performance related to social deprivation. Similar observations were made for socioeconomic status in a German sample ($n = 6,203$)³⁰. Moreover, analyses of data from the USA ($n = 4,592$) on dementia risk detected direct effects of socioeconomic status (defined by educational attainment and income) on cognitive performance and dementia risk as well as indirect effects of lifestyle³¹.

Although it is well known that people with low socioeconomic status have a higher risk for dementia³², socioeconomic status refers only to a person's relative standing in a given societal context. Thus, simply categorizing people as having low or high socioeconomic status fails to capture what specific aspects influence dementia risk. Reviewed evidence of epidemiological studies consistently showed

Box 1 | Established risk factors for dementia

Some modifiable factors have been consistently found to influence dementia risk and are considered well-established dementia risk factors.

Environmental and socioeconomic risk factors

Exposure to air pollution in late life. The reported relative risk for dementia was RR 1.1 (95% confidence interval (CI) 1.1–1.1) for each unit increment of fine particles with a diameter $\leq 2.5 \mu\text{m}$ and $\leq 10 \mu\text{m}$ (ref. 10).

Low education before adulthood. The reported relative risk for dementia was RR 1.6 (95% CI 1.3–2.0) for individuals with low education compared to individuals with high education¹⁰.

Lifestyle risk factors

Current and/or continuing smoking in midlife. The reported relative risk for dementia was RR 1.3 (95% CI 1.2–1.4) for individuals who reported midlife smoking¹⁰.

Excessive alcohol consumption in midlife. The reported relative risk for dementia was RR 1.2 (95% CI 1.0–1.5) for consuming more than 168 g per week of ethanol compared to consuming less¹⁰.

Non-adherence to a Mediterranean-like diet. The reported odds for dementia was 0.89 for individuals who adhere to the Mediterranean diet¹⁶⁰. Hence, non-adherence increased risk.

Physical inactivity in midlife. The reported relative risk for dementia was RR 1.2 (95% CI 1.2–1.3) for physically inactive individuals compared to physically active individuals¹⁰.

Social isolation and/or infrequent social contact in late life. The reported relative risk for dementia was RR 1.6 (95% CI 1.3–1.8) for individuals who were socially isolated (for instance, people who live alone, see their family or friends less than once a month and/or do not participate in weekly group activities)¹⁰.

Lack of cognitive training. Evidence suggests that cognitive training can be beneficial for cognitive performance in old age^{16,17}.

Health-related risk factors

History of traumatic brain injury in midlife. The reported relative risk for dementia was RR 1.7 (95% CI 1.7–1.9) for individuals who had a history of traumatic brain injury compared to those who did not¹⁰.

High blood pressure in midlife. The reported relative risk for dementia was RR 1.2 (95% CI 1.1–1.4) for individuals who had high blood pressure in midlife¹⁰.

Hearing loss in midlife. The reported relative risk for dementia was RR 1.4 (95% CI 1.0–1.9) for every unit increment of 10 dB decrease in hearing¹⁰.

Obesity in midlife. The reported relative risk for dementia was RR 1.3 (95% CI 1.0–1.7) for individuals who were obese in midlife¹⁰.

Type 2 diabetes in midlife. The reported relative risk for dementia was RR 1.7 (95% CI 1.6–1.8) for individual with type 2 diabetes compared to those without¹⁰.

Untreated vision loss and/or blindness. The reported relative risk for dementia was RR 1.5 (95% CI 1.4–1.6) for individuals who had untreated vision loss or blindness such as cataracts or diabetic retinopathy compared to those who did not have these pathologies¹⁰.

High serum LDL cholesterol in midlife. The reported relative risk for dementia was RR 1.3 (95% CI 1.3–1.4) for individuals with serum LDL cholesterol $> 3 \text{ mmol l}^{-1}$ (ref. 10) compared to individuals with lower LDL cholesterol levels.

Depression in midlife. The reported relative risk for dementia was RR 2.2 (95% CI 1.7–3.0) for individuals who had depression compared to those who did not¹⁰.

Sleep problems¹⁷. The reported relative risk for dementia was RR 1.3 (95% CI 1.2–1.4) for individuals with sleep difficulties such as insomnia, fragmentation or daytime dysfunction compared to those who did not have these sleep problems¹⁴.

that financial resources (measured in terms of wealth or income or the lack thereof) are associated with differences in cognitive function in old age. However, it is possible that disadvantages linked to a lack of financial resources, such as neighbourhood characteristics, might specifically contribute to this association.

Neighbourhood characteristics

Neighbourhood characteristics describe the features of the environment in which a person lives, such as demographic composition, proximity to a traffic-bearing road (road proximity), accessibility to services and commercial facilities, and the amount of nature, such as trees, parks and forest, surrounding the residences (greenness).

An increasing number of studies suggest that the neighbourhood in which people live could have a relevant impact on cognitive functioning. A review of 25 studies found modest associations of cognitive performance in older adults with neighbourhood characteristics such

as demographic composition (ethnic, race and age groups associated with diverse aspects of cognitive performance), neighbourhood design (the condition of public spaces or street connectivity) and destination accessibility (distance to community resources or presence of a community centre)³³. Another review of 56 studies found a positive association between cognitive performance in older adults and population density, higher street connectivity (number of streets linked to each individual street in the road network) and/or less street integration (number of turns or choice points to go to access locations), better walkability (ease of getting to places), higher number of public transportation stops, distance from residence to main roads, higher land-use mix (integration of residential, commercial and recreational facilities), more neighbourhood resources (for example, community centres, supermarkets and libraries), more green space (such as park or open grassy areas), and better quality of the neighbourhood built environment (for example, street disrepair index, traffic safety, crime safety or cleanliness)³⁴.

A meta-analysis further confirmed the association between road proximity and an increased risk of cognitive impairment and dementia in older adults³⁵. This result aligned with a UK study of 239,629 participants that found that living in areas of high urban density was a risk factor for dementia and Alzheimer disease³⁶. Indeed, high urban density has been associated with several risk factors for dementia, such as air pollution³⁷. However, living in cities can also have advantages. For example, accessible public transport and easier access to daily life services (such as libraries, post offices and grocery stores) were associated with lower dementia risk in a systematic review³⁸. Overall, there seems to be a complexity of interactions between different neighbourhood characteristics that are not yet fully understood.

Living in a disadvantaged neighbourhood (as measured by indices such as the Area Deprivation Index³⁹) was associated with poorer cognitive performance^{40,41}, faster cognitive decline⁴⁰ and a higher risk for dementia⁴² and Alzheimer disease⁴³. However, individual risk profiles might counterbalance neighbourhood risk. For example, neighbourhood deprivation was no longer associated with dementia risk after adjusting for individual-level factors (including education, social class and chronic illnesses) in a UK study with 2,424 participants⁴⁴.

The associations between green spaces and cognitive performance also suggested complex interactions. The link between tree canopy (tree crowns in the area) and reduced dementia risk was partially mediated by physical activity, social support, sleep duration, diabetes and absence of psychological distress in a study of 109,688 individuals⁴⁵, which suggests that the natural environment might facilitate a healthier lifestyle. However, a meta-analysis of twelve studies found a U-shaped association between neighbourhood greenness and dementia risk, indicating that a medium level of park areas, tree canopies, open grassy areas, and similar green spaces could reduce dementia risk⁴⁶. Analyses of a sample in the USA ($n = 3,047$) indicated that, compared to people living in neighbourhoods with low levels of green space, people living in neighbourhoods with medium levels of green space had a 23% lower dementia risk but people living in neighbourhoods with high levels of greenspace had only an 18% lower dementia risk⁴⁷. High levels of green space in this study might represent rural areas with poor access to services, and therefore the U-shaped association might be driven by the lack of infrastructure⁴⁴.

Overall, evidence suggests that living in a disadvantaged neighbourhood is associated with a higher dementia risk, and neighbourhood characteristics such as proximity to roadways, accessibility to daily life services and green spaces might contribute to this association. However, it is still unclear how the different neighbourhood features are interconnected.

Workplace characteristics

Workplace characteristics are aspects of the daily work environment, such as exposure to magnetic fields and pesticides, cognitive stimulation, work-related stress and shift work.

We did not identify studies that simultaneously assessed these aspects of the workplace. However, epidemiological studies have identified associations between specific workplace characteristics and cognitive functioning in old age.

Exposure to magnetic fields at work was associated with a 26% higher dementia risk in a meta-analysis of five cohort studies and six case-control studies⁴⁸. Exposure to pesticides was associated with an over 90% increased risk of cognitive decline and mild cognitive impairment in a meta-analysis with three studies. However, evidence from two studies on dementia included in this meta-analysis was

inconclusive⁴⁸. A reason for these conflicting findings (besides the very few research results available) might be differences in the type of pesticide. For example, a USA study with 3,084 adults showed a higher risk for Alzheimer disease particularly associated with exposure to organophosphates⁴⁹.

Regarding cognitive stimulation, a review concluded that task complexity at work (specifically tasks involving complex interactions with people or complex handling of data as well as high job control) was associated with a lower dementia risk⁵⁰. Higher demands in information processing and pattern detection (but not in language use) ($n = 2,315$ in Germany)⁵¹ and task variability ($n = 7,003$ in Norway)⁵² were also associated with a lower dementia risk. A multi-cohort study including 107,896 respondents from Europe and the USA further confirmed that a generally high level of cognitive stimulation at work might lower dementia risk even after adjusting for education and health conditions⁵³. Roles involving higher work demands do not necessarily require higher education, nor do they necessarily come with higher income. Therefore, higher work demands seem to contribute to cognitive functioning independently.

Regarding job-related burdens, high job strain (low job control combined with high job demands) was associated with worse cognitive performance and faster cognitive decline among older people who had not retired ($n = 307$ Swedish twins⁵⁴; $n = 1,429$ United States adults⁵⁵). This effect was confirmed by findings from Finland ($n = 1,511$), in which subjectively reported higher levels of midlife work-related stress were associated with a higher risk of mild cognitive impairment, dementia and Alzheimer disease⁵⁶. Interestingly, higher reactivity to stress was associated with dementia risk in a case-control analysis with 2,049 Swedish twins⁵⁷. This finding indicates that inter-individual variability in stress reactivity might be a crucial driver of the negative effect of high job strain and stress on a higher risk for developing dementia in late life.

Night shifts are another workplace characteristic that might influence dementia risk. For example, irregular shift patterns and consistent night shifts were associated with dementia risk in a UK sample of 245,570 adults⁵⁸. A dose-response association between having worked in shifts at night for more years than the other twin and dementia risk was found in two Swedish twin cohorts ($n = 54,482$)⁵⁹. When the authors analysed only the pairs of monozygotic twins (59 discordant pairs), they observed a similar trend⁵⁹. Possible explanations for this non-significant result include the small sample of monozygotic twins working night shifts. Moreover, some moderating factors, such as sleeping problems, were not adjusted for in these analyses and might have influenced the observed effects.

Overall, the epidemiological evidence from data from Europe and the USA indicates that magnetic fields and low cognitive stimulation at work might increase dementia risk. Similar observations were made for pesticides, job stress and shift work, but the number of studies providing evidence is limited.

In sum, the epidemiological evidence suggests that population-level factors that might protect against developing dementia in late life are the availability of sufficient personal financial resources, green spaces in the neighbourhood (but not living in rural areas), accessible daily life amenities and cognitive stimulation at work. By contrast, risk factors that might increase the risk of developing dementia in late life are living close to major roads, living in a disadvantaged neighbourhood, and exposure to magnetic fields at work. Exposure to pesticides at work, job stress and night (shift) work might also have a role, but evidence is limited. Evidence on workplace characteristics

in middle-income and low-income countries is also rare. There might be connections between different population-level risk and protective factors that are still unclear (Table 1, Supplementary Table 1 and Supplementary Figs. 1–3).

Individual-level factors

Individual-level factors are micro-level determinants such as a person's behaviour and health state, including attributes that the person can potentially modify. Examples of modifiable individual-level factors are leisure activities and nutritional style. Individual-level factors that cannot be easily modified are emotional wellbeing and a clinical diagnosis of psychiatric disorders.

Leisure activities

Leisure activities are activities that people carry out in their free time and are often referred to as hobbies.

Although it is well known that cognitive training and physical activities have positive effects on cognitive functioning in old age, other hobbies such as mental activities (for example, playing games or doing puzzles), attending art exhibits or concerts and playing a musical instrument might also have a similar risk-reduction effect. This finding is not surprising, given that these activities involve a cognitive load (they engage attention and working memory⁶⁰), which could be beneficial for the brain even if the person pursues the activity just for fun. Accordingly, it seems reasonable that having many hobbies was associated with a greater dementia risk reduction than having only one hobby, as found in a Japanese study of 22,377 adults⁶¹. Furthermore, an Australian study with 10,318 community-dwelling adults aged 70 and above⁶² reported that pursuing active mental activities (such as playing games or doing crosswords) had a slightly larger protective effect against dementia (9% risk reduction) than passive mental activities (such as watching television or listening to the radio) and artistic activities (such as craftwork or painting; 7% risk reduction for each).

A specific form of leisure activity that is not generally considered a cognitive activity but nonetheless might provide intellectual stimulation includes cultural activities such as attending art or dance exhibits. Receptive arts engagement (for example, going to galleries or museums) had a dose–response effect on memory decline over 10 years in a UK study ($n = 3,911$)⁶³, and active arts engagement (for example, crafting or painting) was related to lower risk of cognitive impairment in a Japanese study ($n = 44,985$)⁶⁴.

Playing a musical instrument was also associated with a decreased risk of dementia, as reviews and meta-analyses published in the past four years indicate^{65,66}. Analyses of Swedish twins ($n = 157$ pairs) showed a 64% lower risk of mild cognitive impairment and dementia associated with playing an instrument⁶⁷. Results from a Japanese study ($n = 52,601$) suggested that practicing karaoke might also be advantageous, especially among women⁶⁸.

This growing body of evidence suggests that various leisure activities might potentially protect cognitive functioning in old age. Further investigation is needed to understand how specific components of these activities (for example, cognitive load, physical requirements and social interactions) contribute to the observed protective effect.

Nutritional style

A person's nutritional style describes the overall choice of foods and beverages they habitually consume. Besides the Mediterranean diet (which is known for its potential to reduce dementia risk), the amount

of ultra-processed food, meat, fruits and vegetables consumed might play a part in cognitive decline and dementia risk.

Ultra-processed foods comprise soft drinks, dairy drinks, powdered mixes for drinks, packet snacks, biscuits, sweets and chocolates, ice creams, packaged bakery items, breakfast cereals and pre-prepared foods⁶⁹. Two reviews have concluded that consuming more of these types of food was associated with a higher risk for dementia^{69,70}. Specifically, high, but not moderate, intake of ultra-processed foods was associated with an increased dementia risk⁷⁰, and higher consumption of processed meat was associated with a higher risk of Alzheimer disease⁶⁹. However, a review focusing specifically on the effect of meat consumption reported no associations with cognitive performance⁷¹, a result that was probably due to considerable heterogeneity between the studies. An analysis of data from 493,888 people in the UK specified an overall 44% higher risk of developing dementia for each additional 25 g per day intake of processed meat; however, the same analysis also showed that the consumption of unprocessed red meat was associated with a reduced dementia risk⁷². The large amount of calories, carbohydrates, saturated fat and sugar contained in ultra-processed foods⁷³ might explain the increased dementia risk.

Fruits and vegetable intake might help to reduce the risk of dementia. A meta-analysis reported an approximately 17% lower dementia risk among people with higher fruit consumption (per increment of 80 g of fruits per day) and a 25% lower dementia risk among people with higher vegetable consumption (per increment of 77 g of vegetables per day)⁷⁴. However, fruit and vegetable consumption were not associated with the risk of Alzheimer disease⁷⁴. Another meta-analysis identified a relationship between each additional 100 g per day of fruit and vegetable consumption and a 13% lower dementia risk⁷⁵. These associations were explained by higher intakes of vitamin A, riboflavin, vitamin C, magnesium, calcium and potassium in a Japanese study with a 24-year follow-up ($n = 1,071$)⁷⁶.

Beverages might also be relevant. Consuming more than three sugar-sweetened soft drinks (one category of ultra-processed foods) a day was not associated with dementia risk in a 10-year risk analysis ($n = 2,888$)⁷⁷, but a 20-year risk analysis revealed a hazard ratio of 2.80 among individuals consuming more than seven servings of sugar in beverages per week ($n = 2,664$)⁷⁸. By contrast, research has shown beneficial effects for coffee consumption⁷⁹. A meta-analysis revealed a U-shaped relationship between the number of cups of coffee a day and risk for Alzheimer disease, in which three to five cups of coffee a day was associated with the lowest risk for Alzheimer disease⁸⁰. Caffeine (a main component of coffee) crosses cell membranes and the blood–brain barrier⁸¹, and therefore could influence memory⁸¹ and attentional control⁸², as experimental studies have shown.

Overall, evidence suggests that nutritional styles with a high intake of ultra-processed foods and sweetened beverages could be a potential risk factor for dementia, whereas nutritional styles with high intakes of fruits and vegetables and a moderate amount of coffee could be a potential protective factor. However, these observations should be interpreted with caution, as evidence is still limited and the nutritional content of any consumed food or beverage might be the driving cause of the observed associations.

Emotional wellbeing

The idea that emotions influence health dates back to Hippocrates⁸³, and evidence has accumulated suggesting that this might be true for dementia risk. Emotional wellbeing has been investigated in the field of dementia in terms of happiness, life satisfaction and having a purpose

Table 1 | Reviewed population-level factors with evidence from original studies

Type of study	Factor	Definition	Sample size	Change in risk	Ref.
Financial resources					
Cross-sectional	Multidimensional poverty	Weighted index of 6 dimensions: education, health, economic activity, living standards, psychological wellbeing and social participation	817	Individuals experiencing multidimensional poverty were 3.88 times more likely to have dementia	25
	Multidimensional poverty	Weighted index of 7 dimensions: education, fair treatment, health, economic activity, living standards, psychological wellbeing and social participation	227	Individuals experiencing multidimensional poverty were 2.31 times more likely to have dementia	24
Prospective longitudinal	Social deprivation	Weighted index of 5 dimensions: education, health insurance status, income, job stability and wealth	11,623	The hazard for dementia was 79% higher for socially deprived individuals	28
	Negative wealth shock	A loss of 75% or more in total wealth over a 2-year period	8,082	The hazard for dementia was 27% higher for individuals with negative wealth shock	26
	High income	Income-to-poverty ratio	3,785	Individuals with high income (having approximately five times more income than the US Census Bureau poverty threshold) had 16% decreased odds of dementia	22
	Wealth	Total value of home, physical wealth, business assets and financial assets	6,346	The hazard for dementia was 42% lower for wealthy individuals (highest tertile)	23
Neighbourhood characteristics					
Cross-sectional	Natural environment	Percentage of green space and private gardens	2,424	Individuals living in environments with the highest quartile of green space and gardens were 2.23 times more likely to develop dementia	44
	Land-use mix	Composite of areas for residential, commercial and recreational uses with various facilities, services and resources	2,424	Individuals living in areas with high land-mix use (highest quartile) had 56% decreased odds of dementia	44
Prospective longitudinal	High urbanicity	Residential density, retail density, public transport and urban centrality	239,629	The hazard for dementia was 21% higher for individuals living in areas with high urbanicity (highest quintile)	36
	Area deprivation	Composite of seventeen indicators (such as income, education, employment and housing) of the area	1,637,484	The hazard for dementia was 17% higher for individuals living in areas with high deprivation (highest quintile)	42
	Tree canopy	Composite of street trees, trees in parks and forests within 1.6 km of home	109,688	Individuals living in areas with more tree canopy ($\geq 20\%$ tree canopy) had 14% decreased odds of dementia	45
	Residential green space	Normalized Difference Vegetation Index reflecting the distance to the nearest park	3,047	The hazard for dementia was about 23% lower for individuals living in areas with medium residential greenness (34th to 66th percentile)	47
Workplace characteristics					
Prospective longitudinal	Night shift work	Job involving night shifts	245,570	The hazard for dementia was 103% higher for individuals who worked night shifts	58
	Low job control	Indicators from the psychosocial job exposure matrix of the Nordic Occupation Classification	913	The hazard for dementia was 70% higher for individuals who had low job control	158
	High routine task intensity	Degree to which the job involved routine manual, cognitive, analytical and interpersonal tasks (routine manual task scale)	7,003	The relative risk for dementia was 166% higher for individuals who did many routine tasks at work (one standard deviation below norm)	52
	High stress reactivity at work	Experiencing symptoms (such as feeling emotionally upset or strain) lasting at least one hour	2,049	Individuals with high stress reactivity were 1.57 times more likely to develop dementia	57
	Midlife work stress	Frequency of struggling to cope with the amount of work and being bothered by constant hurry at work	1,511	Individuals with more midlife work stress (second and third tertiles) were 1.53 times more likely to develop dementia	56
	Pesticides	Job (for 5 years or more) involved pesticide use	3,084	The hazard for dementia was 38% higher for individuals who used pesticides at work	49

Table 1 (continued) | Reviewed population-level factors with evidence from original studies

Type of study	Factor	Definition	Sample size	Change in risk	Ref.
Workplace characteristics (continued)					
Prospective longitudinal (continued)	Shift work	Working shifts	54,482	The hazard for dementia was 36% higher for individuals who worked in shifts	59
	Irregular shift work	Working shifts in current job including never, rarely, sometimes and usually night shift workers	245,570	The hazard for dementia was 19% higher for individuals who worked irregularly in shifts	58
	Information processing at work	Composite of levels of identifying, estimating, evaluating, analysing, decision-making, processing, documenting, updating and communicating information at work	2,315	The hazard for dementia was 11% lower for individuals with high levels of information processing at work	51
	Cognitive stimulation at work	High demands and high control at work	107,896	The hazard for dementia was 18% higher for individuals who had high levels of cognitive stimulation at work	53

Results from studies with meta-analytic evidence are described in the main text, Supplementary Table 1 and Supplementary Figs. 1–3.

in life (for a range of positive emotional states) and in terms of hopelessness, negative thinking and stress including psychological stress exposure over the life course⁸⁴ (for a range of negative emotional states). Regarding positive emotional states, higher positive affect (feeling enthusiastic, attentive, proud or active) and higher life satisfaction were associated with slower memory decline ($n = 991$ in the USA⁸⁵; $n = 8,021$ in South Korea⁸⁶) and a lower dementia risk ($n = 14,286$ in Japan⁸⁷). Having a purpose in life was consistently associated with an over 20% lower risk for dementia in meta-analyses^{88,89}. By contrast, the associations between happiness and cognitive decline were not significant ($n = 17,885$ in the UK⁹⁰; $n = 258$ in the Netherlands⁹¹). Inconsistencies in these findings might be related to negative emotional states that vary between participants but are not usually assessed or analysed simultaneously with positive emotional states.

Techniques such as meditation and mind–body exercises (mindful exercises combining movement sequences, breathing control and attention regulation such as tai chi, qigong or yoga) might have emotional wellbeing benefits associated with better cognitive performance⁹², although the evidence from epidemiological studies is extremely limited. Two studies in the USA demonstrated that meditating at least twice a week ($n = 1,160$)⁹³ and meditating persistently over ten years ($n = 2,168$)⁹⁴ were associated with better cognitive performance over the study periods. The reduction in cognitive decline over the 16-year follow-up was especially relevant among people without depressive symptoms⁹³. Cross-sectional analyses from China showed that practicing tai chi at least once per week was associated with better performance in all sub-scores of the mini mental state examination ($n = 1,732$)⁹⁵ and attention and memory tasks ($n = 42$)⁹⁶.

Negative affect assessed via the positive and negative affect schedule was associated with a higher risk for mild cognitive impairment or dementia in a study with 2,137 women in the USA⁹⁷. Hopelessness in mid-life was associated with a higher risk for Alzheimer disease in a Finnish sample of 2,000 older adults⁹⁸, and repetitive negative thinking was associated with faster cognitive decline and a higher amyloid load in a Canadian sample of 292 older adults⁹⁹.

A review observed that about 80% of the studies on psychological stress showed a meaningful association with dementia risk¹⁰⁰. In particular, high levels of neuroticism and subjectively perceived stress were relevant for memory function among older adults¹⁰¹. The relevance of subjectively perceived stress was further emphasized

by findings from Sweden that showed cumulative effects of frequent subjectively perceived stress on incident dementia ($n = 1,462$)¹⁰² but no effect of the number of negative life events (including areas such as private and working life or deaths of relatives or close friends) or their emotional impact (the person’s interpretation of the event ranging from very positive to very negative) on dementia risk ($n = 2,462$)¹⁰³. Indeed, a review reported inconsistent results for the association between exposure to negative life events and dementia, and concluded that chronic psychological stress might be more relevant to understanding dementia risk than single stressful events¹⁰⁴.

The available evidence suggests that a lack of purpose in life, negative affect (assessed through questionnaires including hopelessness and negative thinking), and chronic psychological stress could be associated with the risk for dementia. The evidence on positive emotional states, such as life satisfaction and emotional states induced via meditation, reflects a similar trend of lower dementia risk, but not all studies report significant findings. Studies using assessment scales that cover the full range from negative to positive affect (instead of using separate instruments for each) would be helpful in exploring interactions between positive and negative emotional states.

Psychiatric disorders

Psychiatric disorders are generally defined as clinically significant disturbances in a person’s cognition, emotional regulation or behaviour¹⁰⁵. Examples of psychiatric disorders are major depressive disorder, bipolar disorder, clinical anxiety, post-traumatic stress disorder and psychotic disorders (including schizophrenia and other mental disorders with thinking and perception alterations). Whereas depression is a well established and widely recognized risk factor for dementia¹⁰, the association between other psychiatric disorders and dementia is not yet established.

Having a history of bipolar disorder was associated with a higher risk for dementia in a meta-analysis of six studies¹⁰⁶. Regarding clinical anxiety, a meta-analysis estimated a 60% higher dementia risk for people diagnosed with this disorder 5 to 20 years before the study¹⁰⁷. By contrast, a selective meta-analysis that used only findings from cohort studies reported an 18% higher dementia risk associated with clinical anxiety earlier in life (mean follow-up 4 to 19 years)¹⁰⁸.

A smaller number of studies have investigated post-traumatic stress disorder. A meta-analysis with four studies reported a pooled

Table 2 | Reviewed individual-level factors with evidence from original studies

Type of study	Factor	Definition	Sample size	Change in risk	Ref.
Leisure activities					
Prospective longitudinal	Creative artistic activities	Engaging in crafts, woodwork, metalwork, painting and/or drawing	10,318	The hazard for dementia was 7% lower for individuals who engaged in creative artistic activities	62
	Passive mental activities	Watching television, listening to music or the radio, and/or reading books, newspapers and magazines	10,318	The hazard for dementia was 7% lower for individuals who engaged in passive mental activities	62
	Active mental activities	Engaging in games, cards, chess, puzzles and/or crosswords	10,318	The hazard for dementia was 9% lower for individuals who engaged in active mental activities	62
	Practicing karaoke	Singing karaoke	52,601	The hazard for dementia was 17% lower for individuals who practiced karaoke	68
	Having a hobby	Having any type of hobby	22,377	The hazard for dementia was 18% lower for individuals who had a hobby	61
	Painting	Creative painting	44,985	The hazard for dementia was 20% lower for individuals who painted as a leisure activity	64
	Crafts	Engaging in crafts	44,985	The hazard for dementia was 29% lower for individuals who engaged in crafts	64
	Playing musical instrument	Playing frequently or occasionally a new and/or familiar musical instrument	157 twin pairs	Twins who played a musical instrument were about 65% less likely to develop dementia than the other twin who did not play an instrument	67
Nutritional style					
Prospective longitudinal	Sugared beverages	Total sugary beverages excluding diet soft drinks	2,664	The hazard for dementia was 180% higher for individuals who consumed a lot of sugar-sweetened beverages (>7 servings per week)	78
	Meat consumption	Processed meat intake such as bacon, ham, sausages, meat pies, kebabs, burgers and chicken nuggets	493,888	The hazard for dementia was 44% higher for each additional 25 g per day intake of processed meat	71
	Unprocessed red meat	Unprocessed red meat (beef, lamb, pork) intake	493,889	The hazard for dementia was 19% lower for each additional 50 g per day increment in unprocessed red meat intake	71
	Sugar-sweetened soft drinks	Intake of high-sugar carbonated beverages such as cola	1,484	The findings were not significant	77
	High fruit consumption	Any fruit intake	1,071	The hazard for dementia was 24% lower for individuals who had a high consumption of fruits (third quartile)	76
	High vegetable consumption	Intake of green and yellow vegetables such as spinach, tomato, pepper, carrot and other vegetables such as cabbage, onion, cucumber and pickles	1,071	The hazard for dementia was 27% lower for individuals who had a high consumption of vegetables (fourth quartile)	76
Emotional wellbeing					
Prospective longitudinal	Frequent stress (three examinations)	Experiencing frequently long periods of stress (such as feelings of irritability, tension, nervousness, fear, anxiety or sleep disturbances) in everyday life	1,462	The hazard for dementia was 251% higher for individuals who had frequent stress	102
	Negative affect	Negative affect items of the Positive and Negative Affect Schedule	2,137	The hazard for dementia was 33% higher for individuals high in negative affect (10-point increase)	97
	Hopelessness	Composite of aiming for impossible-to-reach goals, striving for something, feeling hopeless, and not believing that things change for the better	1,449	Individuals with more hopelessness (median split) were 1.26 times more likely to develop dementia	98
	Frequent stress (one examination)	Experiencing frequently long periods of stress (such as feelings of irritability, tension, nervousness, fear, anxiety or sleep disturbances) in everyday life	1,462	The hazard for dementia was 10% higher for individuals who had frequent stress at only one examination time point	102
	Number of negative life events	Life events such as somatic illness or operation, death of a close relative, or serious illness of a close relative rated as negative	2,462	The findings were not significant	103

Table 2 (continued) | Reviewed individual-level factors with evidence from original studies

Type of study	Factor	Definition	Sample size	Change in risk	Ref.
Emotional wellbeing (continued)					
Prospective longitudinal (continued)	Life satisfaction	Scale of life satisfaction (1 item)	8021	The hazard for dementia was 9% lower for individuals who had high life satisfaction (one standard deviation)	86
	Positive affect (men)	Composite of satisfaction with life, feeling happy, feeling full of energy, feeling that it is wonderful to be alive, and being in good spirits	14,286	The hazard for dementia was 13% lower for men with higher positive affect (for a one-point increment)	87
	Purpose in life	Purpose in life subscale of the Ryff Measures of psychological wellbeing	10,099	The hazard for dementia was 18% lower for individuals who had a higher purpose in life	159
	Positive affect (women)	Composite of satisfaction with life, feeling happy, feeling full of energy, feeling that it is wonderful to be alive, and being in good spirits	14,286	The hazard for dementia was 21% lower for women who had higher positive affect (for a one-point increment)	87

We found only meta-analytic evidence covering the association between psychiatric disorders and dementia risk, and therefore these disorders are not listed in the table. For results from studies with meta-analytic evidence, see Supplementary Table 2 and Supplementary Figs. 1–3.

odds ratio of 2.55, indicating a 155% higher risk for dementia among those who had a diagnosis of post-traumatic stress disorder, that was, however, non-significant¹⁰⁷. Another review that included three population-based cohort studies concluded that having post-traumatic stress disorder was associated with increased dementia risk (hazard ratios ranging from 1.7 to 4.4) in studies with follow-up periods of 11 to 17 years – an effect that was larger for people who made more visits to psychiatric clinics¹⁰⁸.

Meta-analyses reported more than doubled dementia risk for people with a diagnosis of psychotic disorders: pooled RR 2.19 (ref. 108) and pooled RR 2.52 (ref. 109). Risk estimates were higher (pooled RR 3.10) for people with typical age of onset versus people who were older than 60 years old at the onset of psychotic disorder (pooled RR 2.77)¹⁰⁹. Psychiatric symptoms are common among people with dementia¹¹⁰, and some of the people diagnosed with psychotic disorders in older age might already have early symptoms of dementia. Evidence from epidemiologic studies demonstrated that people with bipolar disorder, clinical anxiety, post-traumatic stress disorder and psychotic disorders were more likely to develop dementia than people with no history of psychiatric disorders. The large confidence intervals and heterogeneity across studies should be resolved to advance the interpretation and implementation of the findings.

In sum, epidemiological evidence suggests that individual-level factors might potentially delay dementia onset, as has been shown for leisure activities such as crafts, playing games or a musical instrument, nutritional styles involving a high intake of fruits and vegetables and up to a certain amount of coffee as well as having a purpose in life and meditation. On the other hand, negative affect (such as that induced by chronic psychological stress, hopelessness and repetitive negative thinking), psychiatric disorders and a high intake of ultra-processed foods might be additional risk factors. However, evidence is still scarce, and findings need to be validated (Table 2, Supplementary Table 2 and Supplementary Figs. 1–3).

Interventions and causality

The studies reviewed in the previous sections were observational. Therefore, the links between the explanatory factors and dementia or cognitive decline might not necessarily indicate causality. By contrast, intervention studies can help to verify causal links. Findings from

intervention studies could identify which factors or processes are effective targets for promotion and prevention programmes. Interventions can target either the population (by changing the general conditions in a society) or the individual (by encouraging people to adopt a healthy lifestyle)¹¹¹.

Intervention studies that increase exposure to nature might improve cognitive abilities in older adults. For example, a study in a sample of 25 Belgian employees (mean age 45 years old) found that nature-based activities such as a nature walk, a workshop ‘Land art’, a workshop ‘Bird feeder’, a cycling tour and an ‘Edible nature’ walk (90 minutes, twice a week for three weeks) improved information-processing speed (digit–symbol modalities test, $P < 0.001$) compared to a control group¹¹². Similarly, a horticulture programme with 47 older people in Singapore improved cognitive performance in the mini mental state examination ($P = 0.01$)¹¹³.

Participation in leisure activities, nutritional style, meditation and mind–body exercises can largely be self-determined despite being influenced by population-based factors such as access to resources and environmental constraints. Several individual-level interventions have been evaluated to leverage the modifiability of these factors. For instance, a specialist-led ceramic painting intervention significantly improved performance in the mini mental state examination in a study in Turkey ($n = 15$ in the intervention group, $n = 15$ in the control group; $P = 0.001$)¹¹⁴. Similarly, a study in the USA ($n = 122$) found that active theatre and voice activities over four weeks significantly improved general cognitive performance (combined score of eight cognitive tests $P < 0.001$) and specific performance in immediate word recall ($P < 0.001$), problem-solving ($P < 0.001$), verbal fluency ($P < 0.001$), delayed word recall ($P < 0.05$) and delayed East Boston memory test ($P < 0.05$), but not in the digit span test ($P > 0.05$) or the immediate East Boston memory test ($P > 0.05$)¹¹⁵. Moreover, piano training over multiple months was associated with some cognitive benefits in a Spanish study with 13 participants (benefits in the Stroop test ($P = 0.034$) and Trail Making Test A ($P = 0.022$), but not in the digit span, digit–symbol modalities and lexical task tests¹¹⁶) and in a USA study with 30 participants (interaction effects for condition were $P < 0.015$ for the digit–symbol modalities test and $P < 0.03$ for Trail Making Test B, but there were no associations in the Trail Making Test A, digit span backwards, blocks design and letter–number sequencing tests¹¹⁷).

Improvements in cognitive performance were observed for several types of dietary intervention¹¹⁸, including some programmes that specifically increased fruit and vegetable intake¹¹⁹. For example, results from a crossover trial with 108 healthy adults showed that 16 weeks of consumption of micronized fruits and vegetables compared to a placebo led to improvements in the Stroop test ($P < 0.01$), the Test of the Trail ($P < 0.01$) and a short adaptation of the RIAS memory test ($P < 0.01$)¹²⁰. Most nutritional interventions, however, investigate powders, juices, capsules or specific diets such as the Mediterranean diet, making it difficult to derive effects specifically related to fruits and vegetables.

A meta-analysis focusing on older adults concluded that meditation and mind–body exercises (such as tai chi, qigong or yoga) improved cognitive performance (standardized mean difference was 0.34 for the cognitive tests used in the study), with larger effects for interventions that lasted longer than 12 weeks, were conducted 3 to 7 times a week, or lasted 45 to 60 minutes per session¹²¹. Moreover, a network meta-analysis reported that mind–body exercises and meditation were more effective than physical exercise, non-invasive brain stimulation or cognitive training in improving global cognitive performance in older adults¹²². Overall, evidence suggests that meditation and mind–body exercises benefit cognitive functioning in older age.

Research further suggested that psychosocial interventions can improve emotional wellbeing and cognitive performance concurrently. For example, a 5-week virtual reality intervention with 31 older adults with cognitive impairment improved performance on the Montreal cognitive assessment scale (mild dementia $P < 0.001$, mild cognitive impairment $P = 0.023$), the mini mental state examination (mild dementia $P < 0.001$), and the digit–symbol modalities test (mild dementia $P = 0.009$, mild cognitive impairment $P < 0.001$) compared to before the intervention¹²³. Similarly, a 6-week cognitive stimulation intervention improved cognitive performance (mini mental state examination, $P < 0.001$) among older adults in the intervention group ($n = 40$) compared to the control group ($n = 40$)¹²⁴.

Multi-domain interventions that simultaneously target several health and lifestyle factors (including nutritional style, physical activity, cognitive training, social activity and management of metabolic and vascular risk factors) might enhance cognitive functioning¹²⁵. However, a Cochrane review concluded that these multi-domain interventions have only a small effect on cognitive functioning (mean difference was 0.03 in composite Z-scores of a neuropsychological test battery; mean difference was 0.76 in the Montreal cognitive assessment scale), and there was no evidence for reduced dementia incidence associated with these interventions¹²⁶. Addressing additional risk and protective factors could increase the effects of multi-domain trials. However, targeting more domains is not always more effective. Tailored strategies need to be developed to address a person's most relevant risk factors.

Potential mechanisms

Several studies provide some indications about the mechanisms and pathways through which population-level and individual-level factors influence the risk of cognitive decline and dementia. For instance, some factors might lead to structural differences in the brain. High income¹²⁷, high levels of mental demands at work¹²⁸, and practicing meditation¹²⁹ or yoga¹³⁰ have all been associated with greater grey matter volume in the brain. Specifically, yoga training¹³¹ has been associated with greater volume in the hippocampus – a brain area that is altered early in Alzheimer disease. Having a higher income¹³², practising meditation¹³³, singing in a choir¹³⁴ and participating in musical activities have been

linked to better brain connectivity¹³⁵. By contrast, neighbourhood deprivation¹³⁶, low job demands¹³⁷ and longer years of shift work with sleeping problems¹³⁸ have been associated with smaller grey matter volume. Also, higher levels of worry and rumination¹³⁹ as well as emotional distress¹⁴⁰ were associated with accelerated brain ageing. Living in a deprived neighbourhood was related to more cortical thinning in Alzheimer disease signature regions¹⁴¹, more neurofibrillary tangles and higher neuritic plaque staging¹⁴².

Epigenetics (the regulation of gene expression induced by exposure to certain environmental factors)¹⁴³ might also influence inter-individual differences in the brain linked to higher dementia risk. Although evidence is still scarce, observations indicate associations between risk factors and DNA methylation biomarkers that determine an individual's biological age using biological tissue. For example, more greenness in the neighbourhood was associated with slower biological ageing, as measured using DNA methylation parameters^{144,145}. Some high-risk factors, such as long-lasting exposure to strong magnetic fields¹⁴⁶, might even cause nuclear DNA changes.

Inflammatory and endocrine changes might also moderate the effect of modifiable lifestyle factors on dementia risk. Indeed, endocrine changes might be the reason why psychological stress increases dementia risk. Being exposed to prolonged psychological stressors, financial insecurity, worrying or repetitive negative thinking, living in a disadvantaged neighbourhood, and having a hazardous workplace (among other risk factors) might negatively influence memory abilities – the core symptom of dementia – and the hippocampus^{147,148}. By activating stress-response genes¹⁴³, stress can also lead to epigenetic changes in the brain¹⁴⁹ that might influence brain health over time. Complementarily, organophosphate pesticides can trigger inflammatory responses that could lead to Alzheimer disease pathology¹⁵⁰. By contrast, meditation might lead to decreases in high-sensitivity c-reactive protein and interleukin (IL–6 and IL–1 β) levels¹⁵¹. Meditation has also been associated with positive changes in the hypothalamic–pituitary–adrenal axis and renin–angiotensin–aldosterone system hormone levels¹⁵². Further studies are needed to specify these neurobiological effects.

Summary and future directions

Our Review provides evidence for factors that are not yet widely acknowledged and for which research is emerging that supports their contribution to the risk of cognitive decline and dementia in older adults (Fig. 1). The number of research studies investigating population-level and individual-level factors on dementia risk has increased immensely over the past decades. The fact that meta-analytic data were not available for all of the reviewed factors (such as financial insecurity, types of leisure activity, psychological stress or meditation) highlights that more epidemiological studies are still needed. Similarly, other lifestyle factors that are not covered in this Review might be identified soon. These insights will lead to a more comprehensive understanding of which and how less well-established lifestyle factors might influence dementia risk and the conditions under which cognitive health can thrive.

According to the biopsychosocial framework, the social and environmental context in which a person is embedded influences psychological and behavioural characteristics, which determine health in interaction with individual characteristics¹⁵³. We agree that the individual-level and population-level factors are often interconnected. For instance, financial resources and the neighbourhood built environment influence individual choices. As a study observed, people subject to compulsory military re-assignment increased their physical activity when fitness and recreation facilities were closer to them¹⁵⁴.

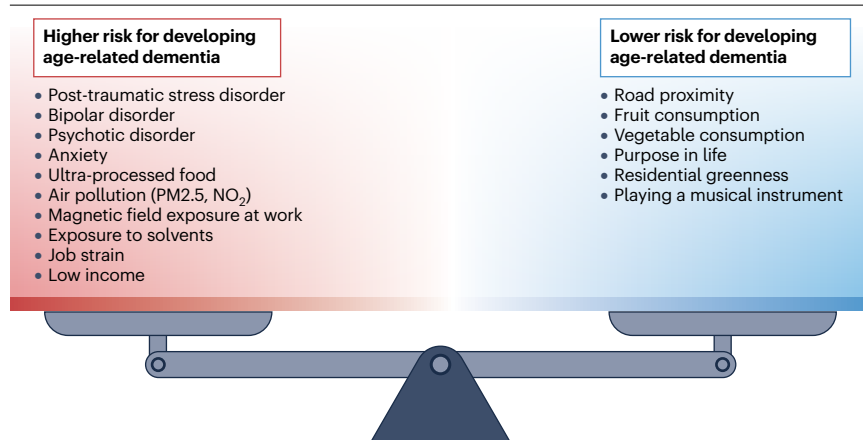


Fig. 1 | Less well-established dementia risk and protective factors with meta-analytic evidence. The effect estimates of meta-analyses (Supplementary Figs. 1–3) reported in the reviewed literature suggest that some of the identified population-level and individual-level factors might confer higher risk for dementia in older adults (left, red), whereas other identified factors might be protective (right, blue). NO₂, nitrogen dioxide; PM2.5, fine particulate matter with a diameter of 2.5 µm or less.

Future studies should investigate interactions between risk factors and across levels, and the extent to which these interactions shape dementia risk. Understanding these relationships will guide the optimal development of policy-level interventions – including access to appropriate health and support services and positive lifestyle opportunities.

As the understanding of risk factors grows, translating this knowledge into effective population-level and individual-level interventions remains a key challenge. Although researchers do not yet fully understand the pathways involved, the increasing number of people with dementia worldwide underscores the urgency of prioritizing intervention research aimed at achieving tangible improvements for people at risk. Population-level interventions could comprise policies aimed at improving neighbourhoods, work conditions, education opportunities or financial security and that might help to modify individual behaviours indirectly and at scale¹⁵⁵. Some examples of such population-level interventions are the adaptation of taxes (such as taxation of sugar-sweetened beverages), adding or removing fees, reducing or increasing availability, marketing or bans (such as marketing bans for cigarettes), labelling (for example, food package labelling), and providing adequate equipment for the general population¹¹¹.

Brain health education should be promoted to enhance knowledge about risk and protective factors across all ages¹¹¹. Simultaneous small initiatives from communities and interested groups could also make a difference. For instance, individual-level interventions could be developed to target the motivation to participate in leisure activities, find a purpose in life, reduce stress or improve coping skills. Variation of specific components of such individual interventions (social embeddedness, physical requirements and cognitive load) might lead to insight into which aspects contribute most to protective effects. Digital innovations, such as computerized cognitive training and mindfulness programmes, might be a practical solution because they can be individualized and integrated into workplace and community programmes, and can save intervention delivery costs.

Population-level approaches aim to create environments that make healthy lifestyle choices the default option. The development of promotion and prevention programmes should consider geographical and cultural differences across citizens and respect the interests and personal values of the targeted groups as well as their individual autonomy. Ethical reflection is crucial in establishing collaborations between policymakers, researchers and communities involved in designing, testing and implementing such preventative measures¹⁵⁵.

The burden of individual interventions should not outweigh their potential benefits, and stigmatization or medicalization of healthy or at-risk adults must be avoided¹⁵⁶. This consideration is particularly relevant for individual-level factors that cannot be easily influenced, such as psychiatric disorders. Managing symptoms and improving treatments must be the focus of interventions that aim to lower dementia risk in people with these conditions, which requires investigation. Complementarily, emotional distress, including negative affect, is a common symptom of psychiatric disorders¹⁰⁵. Moreover, stressful life experiences have been aetiologically associated with the onset of psychiatric disorders¹⁵⁷. Addressing emotional distress could compensate for the negative emotional impact of psychiatric symptoms or psychological stress in people with psychiatric disorders.

The risk and protective factors identified in our Review show promise as targets for preventative measures and warrant further research. A more comprehensive understanding of how these factors act and interact, particularly in underserved communities, will identify synergies or trade-offs that maximize the combined benefits of promotion and prevention strategies, enable the development of personalized interventions and foster societal changes that support sustainable lifestyle practices to improve cognitive ageing.

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Author contributions

F.S.R. and L.M.H. researched data for the article. All authors contributed substantially to discussion of the content, wrote the article and reviewed and/or edited the manuscript before submission.

Competing interests

The authors declare no competing interests.

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